NOT MEASUREMENT SENSITIVE

MIL-M-49504(TM)
23 JANUARY 1991

MILITARY SPECIFICATION

MANUALS, TECHNICAL: COMBAT VEHICLE PRECOMBAT CHECKLIST

This specification is approved for use by the Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 **Purpose**. This specification provides the detailed requirements for the preparation of an operator's, pocket size, precombat checklist (PCL). The PCL includes before operation, mission-essential, maintenance/operation checks to be performed before combat to assure equipment is mission capable.

1.2 **Examples/figures**. The figures used in this specification are examples only. The text of this document takes precedence over the figures.

1.3 **Applicability**. Only combat vehicles will have PCLs. Each item specified herein will be included in the PCL unless otherwise specified by the contracting activity. Each time a combat vehicle operator's technical manual (TM) undergoes a change or revision that involves the item numbers and/or procedures in the PMCS table(s) from which the PCL was taken, the PCL must be changed/ revised.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: USAMC Materiel Readiness Support Activity, ATTN: AMXMD-MP, Lexington, KY 40511-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC A6016

AREA TMSS

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2. APPLICABLE DOCUMENTS

2.1 **Government documents**.

2.1.1 **Specifications, standards, and handbooks**. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS), and its supplement, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-38784	Manuals, Technical: General Style and Format Requirements
MIL-M-85337	Manuals, Technical: Quality Assurance Program, Requirements for

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents**. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DOD Directive 5320.24	Distribution Statements on Technical Documents

(Copies of DOD Directive 5320.24 can be obtained from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 **Non-Government publications**. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the document, which is DOD adopted, shall be that listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issue of a

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document not listed in the DODISS shall be the issue of the non-government document which is current on the date of the solicitation.

ASTM D3951 (DOD adopted)	Standard Practice for Commercial Packaging

(DOD activities can obtain copies of ASTM D3951 from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094. Other Government activities, contractors, and other private concerns must procure copies from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 **Order of precedence**. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General.

3.1.1 **Accuracy and ease of use**. The PCL shall have accurate technical content and be easy for the operator or crew to use.

3.1.2 **Level of coverage**. The PCL shall contain only those before operation tasks necessary to assure that the equipment/system is ready for combat.

3.1.3 **Illustrations**. Illustrations shall not be used in the PCL.

3.2 Style and format requirements.

3.2.1 **General**. The general style and format of the PCL shall be in accordance with MIL-M-38784.

3.2.2 **Content**. The PCL shall be one sheet front and back, and shall consist of a front cover (figure 1) and the checklist (figure 2).

3.2.3 **Manual size**. The PCL shall have a trim size of 5 by 8 inches and a maximum printed area of 4 3/4 by 7 3/4 inches. The maximum printed area shall include all printed matter, including publication and page numbers.

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3.2.4 **Lamination and printing**. Each card of the PCL shall be printed on white 110 index cardstock and shall be laminated on both sides with .005 inch thick clear polyester with a minimum overall thickness of .019 inches. The PCL shall have a 3/8 inch hole punched in the upper left hand corner, 5/16 inch from hole center to top and side of card.

3.2.5 **Type size**. Type size shall not be smaller than 8 point.

3.2.6 **Page numbering**. The checklist of the PCL shall be numbered arabic numeral 2. Page one, the cover, shall not be numbered.

3.3 **Technical content**.

3.3.1 **Cover sheet**. The cover sheet shall consist of the following in the order given:

a. Publication number.

b. Title block.

c. Warning and note.

d. Distribution statement.

e. Authentication block.

f. Special distribution instructions.

g. Military service.

h. Publication date.

3.3.1.1 **Publication number**. The publication number of a PCL shall be the same basic TM identification number as the operator's manual with a -PCL in place of -10. The publication number shall be placed in the upper right hand corner of the front cover as shown in figure 1.

3.3.1.2 **Title block**. The title block of the PCL shall consist of the system/equipment type (e.g., combat vehicle), the words "PRECOMBAT CHECKLIST FOR:", and the name of the system/equipment. See figure 1.

3.3.1.3 **Warning and note**. The cover sheet shall have the following warning and note verbatim:

"WARNING

THIS CHECKLIST IS TO BE USED ONLY WHEN AUTHORIZED BY THE COMMANDER. IT IS NOT INTENDED TO BE USED IN PLACE OF THE PMCS."

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"NOTE

THIS CHECKLIST CONTAINS CHECKS TO BE PERFORMED BEFORE COMBAT TO ASSURE VEHICLE IS MISSION CAPABLE. THE CHECKLIST SEQUENCE NUMBERS ANT THE TABLE NUMBERS HEREIN CORRESPOND TO THOSE PMCS CHECKS IN (INSERT OPERATOR TECHNICAL MANUAL NUMBER). REPORT CHECK RESULTS TO YOUR IMMEDIATE LEADER."

The above shall be placed on the cover as shown on figure 1.

3.3.1.4 **Distribution statement**. The distribution statement shall be in accordance with DOD Directive 5320.24.

3.3.1.5 **Authentication block**. The authentication block shall be added by the Government and shall be positioned on the cover sheet as shown on figure 1.

3.3.1.6 **Special distribution instructions**. Any special distribution instructions shall be placed on the cover as shown on figure 1.

3.3.1.7 **Military service**. The proponent military service (e.g., Department of the Army) shall be centered below the special distribution instructions. See figure 1 for example.

3.3.1.8 **Date**. The date of the PCL shall be centered below the military service as shown on figure 1.

3.3.2 **Checklist**.

3.3.2.1 **Format**. The checklist shall be formatted depending on what the preventive maintenance checks and services (PMCS) table/ operator's manual looks like. If there is one PMCS table, the checks shall be listed without a header. See figure 2. If there is more than one PMCS table, the checks shall be listed by table number. See figure 3. If the operator's manual has more than one volume, the checks shall be listed by volume number (e.g., 10-1, 10-2). See figure 4. The checks shall have the same numbers as checks in the PMCS table from which they were selected. Each check shall consist of a title (e.g., vehicle exterior) printed in all capital letters and a brief explanation/ summary of the check, printed in capital and small letters. See figures 2, 3, and 4 for examples of checks.

3.3.2.2 **Criteria for selection of checks**. The checks to be included in the combat vehicle PCLs shall be tasks from the operator's PMCS table. These checks shall generally be, but are not limited to, before operation, mission-essential, maintenance/ operation checks. The checks shall be in a logical sequence, and

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the crew members shall be able to perform all checks in the checklist in 20 minutes or less. The checks selected shall be those checks determined to be mission-essential based on the Requirements for Operational Capability (ROC), the Mission Needs Statement (MNS), and/or other requirements (e.g., safety). The checklist shall consist of the minimum essential checks required to be sure fire control, armament, communications, and automotive equipment are functioning and the system/end item can perform its primary mission (i.e., move, shoot, and communicate).

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection**. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and devices conform to prescribed requirements.

4.1.1 **Responsibility for compliance**. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 **Quality assurance provisions**. The contractor shall be responsible for quality assurance in the preparation of the PCL. Quality assurance provisions shall be in accordance with MIL-M-85337 to the extent specified by the contracting activity.

4.3 **Contracting activity inspection**. Material furnished by the contractor as being in accordance with this specification shall be subject to inspection, and approval or disapproval, by the contracting activity as specified by the terms of the contract. Inspection will be performed by the contracting activity prior to acceptance.

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4.4 **Validation**. When specified by the contracting activity, the contractor shall validate the PCL in accordance with the requirements of MIL-M-85337. Unless otherwise specified by the contracting activity, the contractor shall validate the PCL tasks by actual performance. The contracting activity reserves the right to witness the validation process.

4.4.1 **Validation certification**. When specified by the contracting activity, the contractor shall prepare a validation certificate attesting to the accuracy and adequacy of the PCL (see 6.3).

4.4.2 **Verification**. When specified by the contracting activity, representatives of the contracting activity will verify the contractor's validation. The contracting activity reserves the right to perform verification by one of the methods listed in a through d below. The methods listed are in addition to reviewing the PCL for conformance to the requirements of the governing documents and inspection of material.

a. Performing 100 percent of the tasks included in the PCL by using military personnel of the type and qualifications of those expected to perform the tasks when the PCL is used. Tasks must be performed successfully at least once. All performances will be monitored by a Government subject-matter expert, and a master copy of the PCL being verified will be maintained to record all discrepancies. All verified material will be updated and given a final desk review. When resources and time constraints limit the feasibility of performing a contractor validation and a Government verification as separate entities, these requirements may be melded into one, subject to the approval of the contracting activity. This option must have agreement of both the materiel developer and the user.

b. Reviewing the technical content of the PCL by having Government personnel perform tasks included in the PCL specified by the contracting activity on the equipment, either at the contractor's plant or at Government facilities.

c. Performing the contractor's validation at the contractor's plant, as scheduled by the contractor in accordance with the validation schedule.

d. Witnessing the validation reviews performed by the contractor.

4.5 **Task corrections**. If the PCL does not meet the inspection requirements above, the contractor shall correct it until it meets those requirements.

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4.6 **Government-furnished information**. The contractor shall accurately and properly include all government-furnished information in the PCL.

5. PACKAGING

5.1 **Packaging requirements**. Packaging for delivery shall be in accordance with the requirements of ASTM D3951.

5.2 **Marking requirements**. Packages shall be marked in accordance with MIL-STD-129 and ASTM D3951, as applicable, and shall include the publication number and date.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful. It is not mandatory.)

6.1 **Intended use**. This specification covers the preparation of operator's PCLs.

6.2 **Acquisition requirements**. Acquisition documents should specify the following items:

a. Title, number, and date of this specification.

b. Title and number of the TM (PCL).

c. Source of validated and verified PMCS tables for use in preparing the PCL.

d. Requirements for validation and verification.

e. The issue of the DODISS cited or, if required, the specific issue of the referenced documents.

6.3 **Data requirements**. The following data item description (DID) should be considered when this specification is cited. Also, it should be listed on a Contract Data Requirements List (DD Form 1423), if required, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

DID number	DID Title
DI-M-2196	Manual, Technical: Validation Certification

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The above DID was cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be researched to ensure that only current and cleared DIDs are cited on the DD Form 1423.

6.4 **Technical manual acquisition**. This specification must be listed on the Contract Data Requirements List (DD Form 1423) in order to acquire the TM described by this specification, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

6.5 **Subject term (key word) listing**.

Mission-essential
PCL
Pocketsize

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TM 9-2350-255-PCL

COMBAT VEHICLE
PRE-COMBAT CHECKLIST FOR:
TANK, COMBAT, FULL TRACKED, 105MM
GUN, M1, IPM1, GENERAL ABRAMS

(NSN 2350-01-061-2445)
(NSN 2350-01-138-8738)

WARNING

THIS CHECKLIST IS TO BE USED ONLY WHEN AUTHORIZED BY THE COMMANDER. IT IS NOT INTENDED TO BE USED IN PLACE OF THE PMCS.

NOTE

THIS CHECKLIST CONTAINS CHECKS TO BE PERFORMED BEFORE COMBAT TO ASSURE VEHICLE IS MISSION CAPABLE. THE CHECKLIST SEQUENCE NUMBERS HEREIN CORRESPOND TO THOSE PMCS CHECKS IN TM 9-2350-255-10-1. REPORT CHECK RESULTS TO YOUR IMMEDIATE LEADER.

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By Order of the Secretary of the Army:

CARL E. VUONO
General, United States Army
Chief of Staff

Official:

THOMAS F. SIKORA
Brigadier General, United States Army
The Adjutant General

Special Distribution:

To be distributed in accordance with DA Form 12-37, Block 1088, four per account, Operator Maintenance Requirements for Tank, Combat, Full-Tracked, 105MM, M1 Abrams.

DEPARTMENT OF THE ARMY
8 MARCH 1990

FIGURE 1. Example of front cover.

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1,4.	**VEHICLE EXTERIOR:** Walk Around And Check For Fluid Leaks, Obvious Damage, Tampering, And Missing Parts That Prevent Operation.
5.	**ENGINE COMPARTMENT:** Check Inside Engine Compartment For Leaks And Damage.
20.	**ENGINE OIL CHECK:** Check Oil Level.
14.	**COOLANT LEVEL:** Check Coolant Level.
2, 16-17.	**FIRE EXTINGUISHERS:** Check Seals And Bottle Pressures.
40-46.	**ENGINE INSTRUMENTS:** Start Engine, Check Instruments.
37.	**COMMUNICATION SYSTEM:** Radio And Intercom Operational.
34,56, 60.	**GUNNER CONTROL HANDLES:** Gun Elevation And Turret Traverse Operational.
	FIRE CONTROL SYSTEM:
28.	A. Nightsight Check.
12,59.	B. M60 Gun Check.
	C. TOW System Self-Test.

2

FIGURE 2. **Example of a checklist for a manual with one PMCS table.**

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TABLE 2-1

1. **VEHICLE EXTERIOR:** Walk Around And Check For Fluid Leaks, Obvious Damage, Tampering, And Missing Parts That Prevent Operation.

21 & 40. **FIRE EXTINGUISHER SYSTEM:** Check Bottle Pressure.

43. **ENGINE/INSTRUMENTS:** Start Engine, Check Instruments.

****NOTE: TURN TIS TO STANDBY MODE*****

24. **COMMUNICATION SYSTEM:** Radio And Intercom Operational.

TABLE 2-2

1,2 & 12. **MACHINE GUNS:** Mounted And Operational.

6. **TURRET AMMUNITION STOWAGE:** Compartment Doors Operational.

9. **COMMANDER/GUNNER POWER CONTROL HANDLES:**
   A. Gun Elevation/Turret Traverse.
   B. Aux Hyd Gauge Reads 1150-1700.
   C. Electrical Triggers And Blasting Device Operational.

13. **FIRE CONTROL SYSTEM:**
   A. TIS Check Out.
   B. Computer Self-Test.
   C. Computer Data Check.
   D. GPS Function Check.

4.1. **STUB BASE DEFLECTOR INSTALLED.**

2

FIGURE 3. **Example of a checklist for a manual with more than one PMCS table.**

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TM 9-2350-284-10-1

2,3. **VEHICLE EXTERIOR:** Walk Around And Check For Fluid Leaks, Obvious Damage, Tampering, And Missing Parts That Prevent Operation.

******NOTE: LOWER RAMP******

4. **HULL DRAIN PLUG FRONT:** Close Front Drain Plug.

12. A,B. **FIRE EXTINGUISHER SYSTEM FRONT:** Check Bottle Pressure.

15-17. **FIRE EXTINGUISHER SYSTEM REAR:** Check Bottle Pressure.

4,15. **HULL DRAIN PLUG REAR:** Close Rear Drain Plug.

7-8. **ENGINE/INSTRUMENTS:** Start Engine, Check Instruments.

TM 9-2350-284-10-2

******POWER UP ISU******

13 A,B. **COMMUNICATION SYSTEM:** Radio And Intercom Operational.

23. **NBC SYSTEM CHECK.**

17A-B, 26. **COMMANDER/GUNNER CONTROL HANDLES:** Gun Elevation And Turret Traverse Operational.

FIRE CONTROL SYSTEM:

29,30. A. ISU Checkout.
33. B. 25MM Gun Checkout.
35. C. Coaxial Machine Gun Function Check.
D. TOW System Self-Test.

2

FIGURE 4. **Example of a checklist for a manual with more than one volume.**

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Custodian: Army - TM

Review Activities:

User Activities:

Preparing Activity: Army - TM

Project Number: TMSS A265

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Commander
USAMC Materiel Readiness Support Activity
ATTN: AMXMD-MP
Lexington, KY 40511-5101

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-49504(TM)	2. DOCUMENT DATE (YYMMDD) 910123

3. DOCUMENT TITLE
Combat Vehicle Precombat Checklist

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME USAMC Materiel Readiness Support Activity	b. TELEPHONE (Include Area Code) (1) Commercial (606) 293-3469	(2) AUTOVON 745-3469
c. ADDRESS (Include Zip Code) ATTN: AMXMD-MP (TMSS)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office	

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